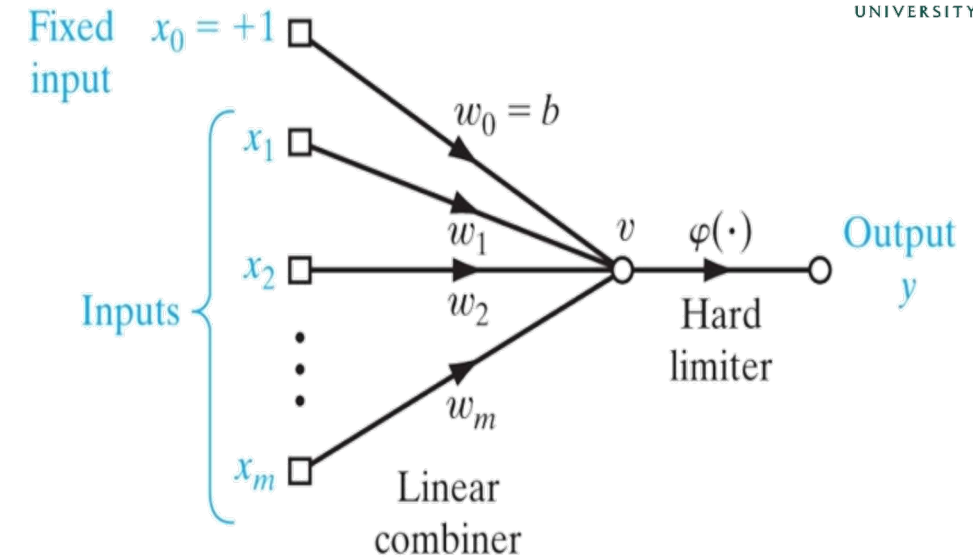


Single-Layer Perceptron & Bayes Classifier

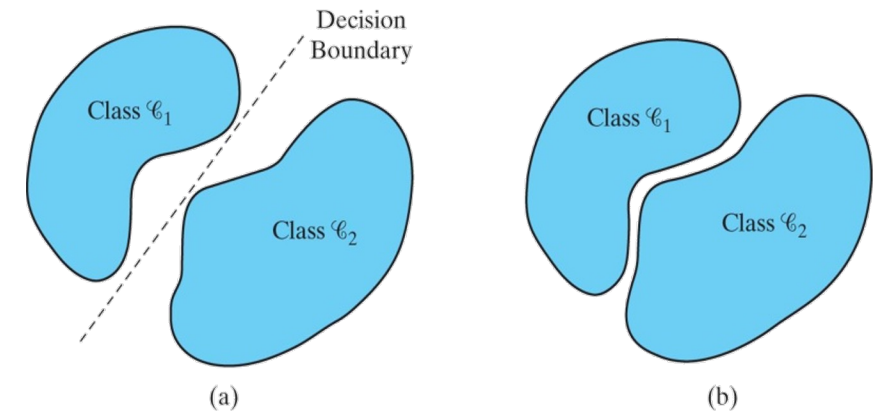
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Introduction

- Resonblatt's [1958] **perceptron** is a neural network used for classification of **linearly separable** patterns
- Consists of a single layer of neurons
- **Perceptron Convergence Theorem:**
 - If patterns are linearly separable, converges and positions the decision surface in the form of a “hyperplane” between the two classes
 - Can be extended for multiple classes
 - One output neuron per class



Perceptron Signal-Flow Graph

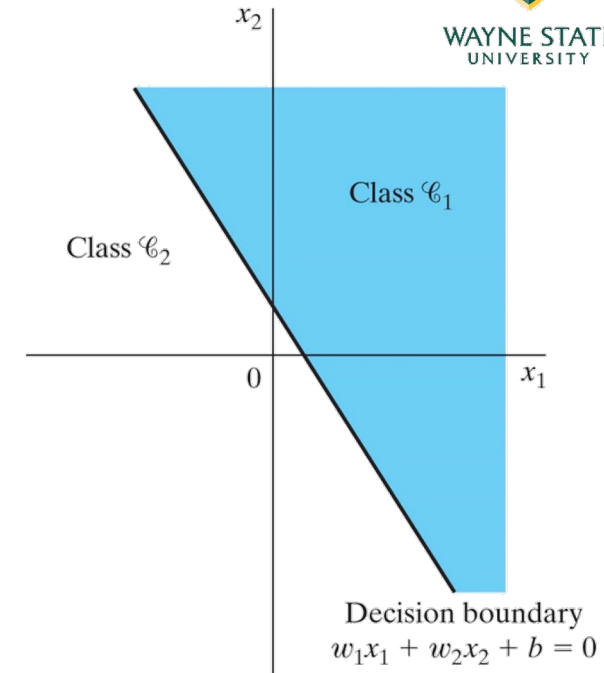


Linearly Separable & Unseparable Classes

Perceptron

- Employs threshold transfer function (outputs ± 1)
- **Goal:** Correctly classify points $x_1, x_2, \dots, x_m$ into one of two **linearly separable** classes C_1 or C_2
- **Decision:** Assign input to class based on output
- **Extensible to Multiple Classes:** Assign input to neuron with largest activation potential (v)
- Neuron bias allows boundary to shift from origin
- **Iterative Process:** Present data points in random order
- Weights updated using **error-correction rule**:

$$w(n + 1) = w(n) + \eta[d(n) - y(n)]x(n)$$



Decision Boundary Hyperplane
(straight line for
2-D two-class problem)

Perceptron Algorithm

Variables and Parameters:

$\mathbf{x}(n)$ = $(m + 1)$ -by-1 input vector
 $= [+1, x_1(n), x_2(n), \dots, x_m(n)]^T$

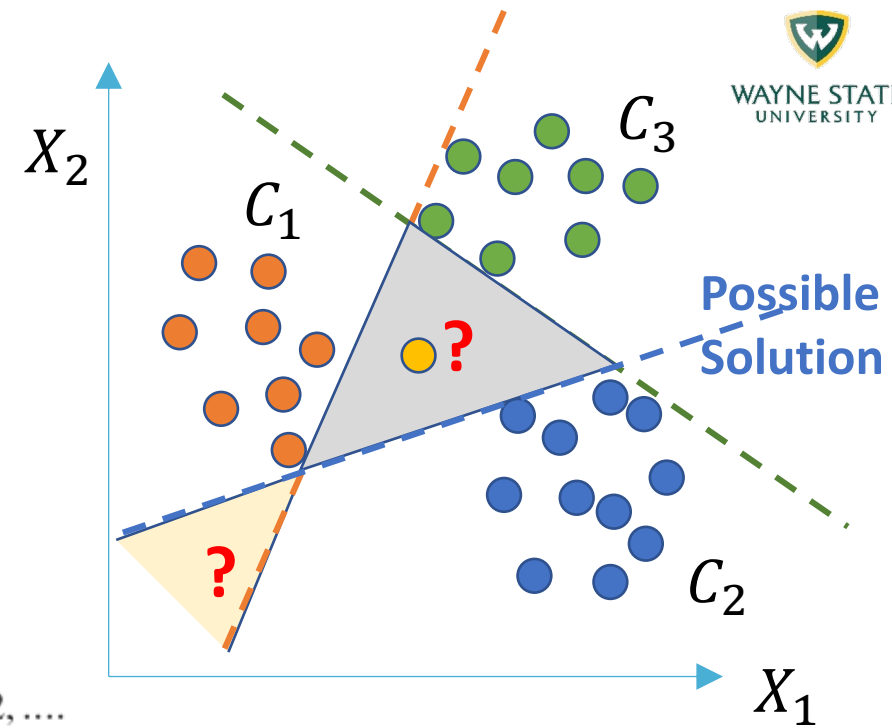
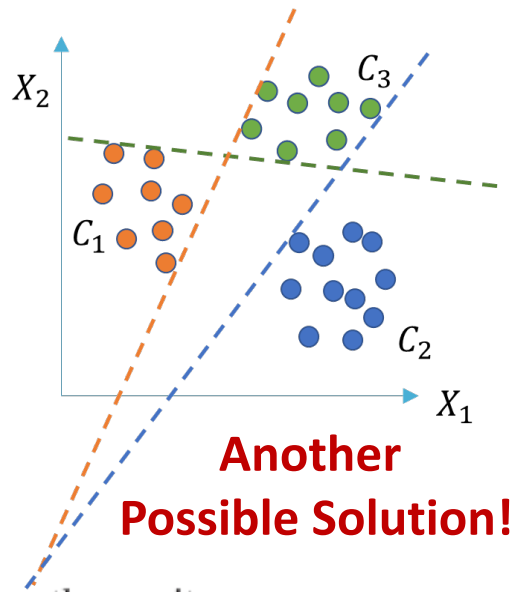
$\mathbf{w}(n)$ = $(m + 1)$ -by-1 weight vector
 $= [b, w_1(n), w_2(n), \dots, w_m(n)]^T$

b = bias

$y(n)$ = actual response (quantized)

$d(n)$ = desired response

η = learning-rate parameter, a positive constant less than unity



1. *Initialization.* Set $\mathbf{w}(0) = \mathbf{0}$. Then perform the following computations for time-step $n = 1, 2, \dots$
2. *Activation.* At time-step n , activate the perceptron by applying continuous-valued input vector $\mathbf{x}(n)$ and desired response $d(n)$.
3. *Computation of Actual Response.* Compute the actual response of the perceptron as

$$y(n) = \text{sgn}[\mathbf{w}^T(n)\mathbf{x}(n)]$$

where $\text{sgn}(\cdot)$ is the signum function.

4. *Adaptation of Weight Vector.* Update the weight vector of the perceptron to obtain

$$\mathbf{w}(n + 1) = \mathbf{w}(n) + \eta[d(n) - y(n)]\mathbf{x}(n)$$

where

$$d(n) = \begin{cases} +1 & \text{if } \mathbf{x}(n) \text{ belongs to class } \mathcal{C}_1 \\ -1 & \text{if } \mathbf{x}(n) \text{ belongs to class } \mathcal{C}_2 \end{cases}$$

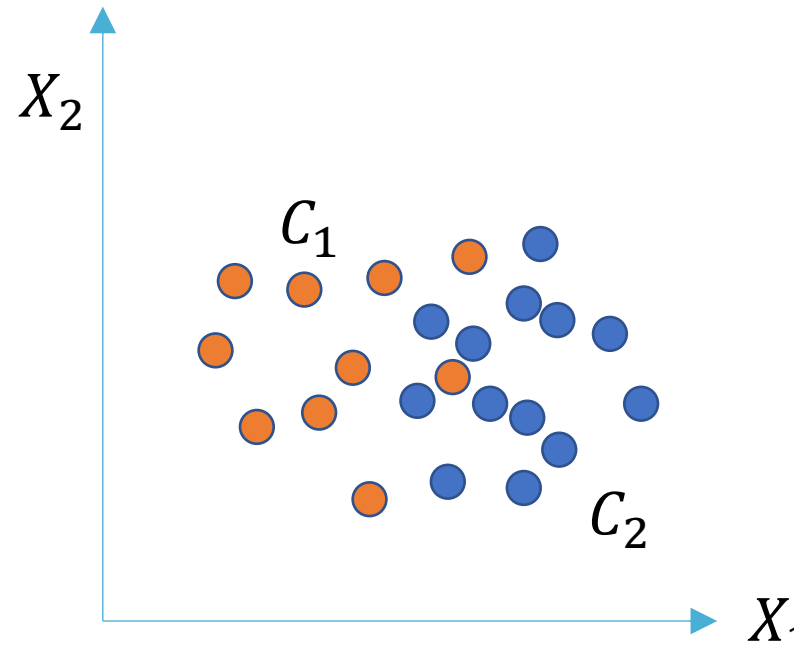
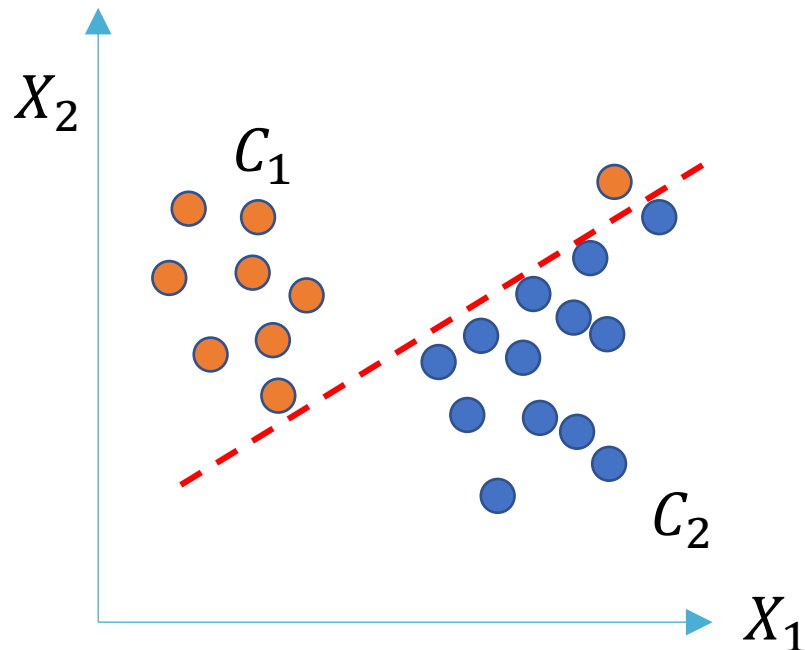
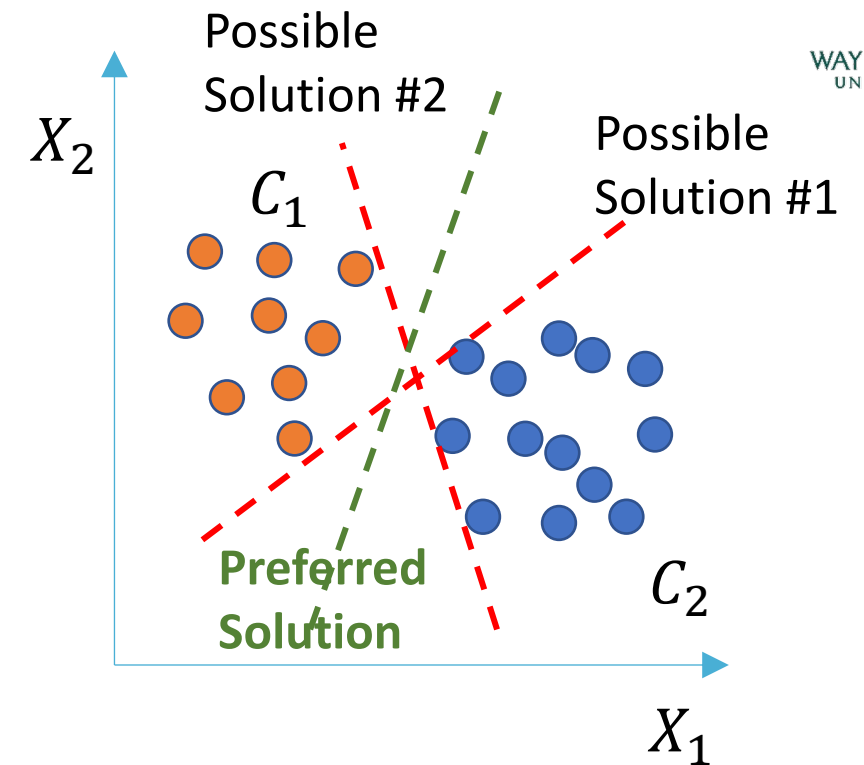
5. *Continuation.* Increment time step n by one and go back to step 2.

Termination Criteria?

Learning Rate?

Perceptron: Limitations

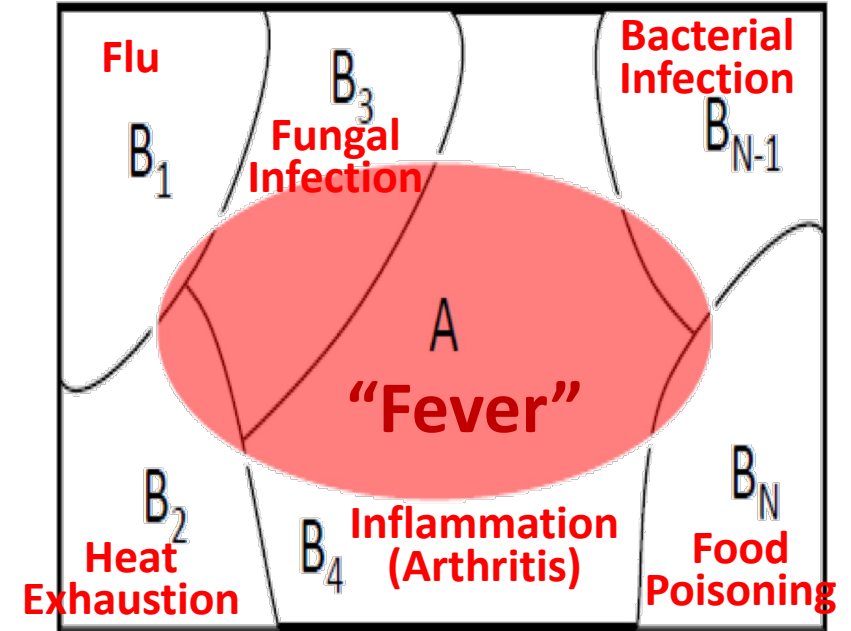
- Optimality of Decision Surface!
 - Stops learning once data points are correctly classified
- Robustness to Noise
- What about Non-Separable Classes?



Bayes Theorem

- Assume $\{B_1, B_2 \dots B_N\}$ is a partition of S
- Suppose that event A occurs
- What is the probability of event B_j ?
- **Bayes Theorem/Rule:** From definition of conditional probability and Theorem of total probability:

$$\begin{aligned}
 P[B_j|A] &= \frac{P[A \cap B_j]}{P[A]} \\
 &= \frac{P[A|B_j]P[B_j]}{\sum_{k=1}^N P[A|B_k]P[B_k]}
 \end{aligned}$$



Bayes Classifier

- Has some resemblance to the Perceptron
 - Under Gaussian distributions, Bayes classifier reduces to a linear classifier
- When used for pattern classification:

$$P[\omega_j|x] = \frac{P[x|\omega_j]P[\omega_j]}{\sum_{k=1}^N P[x|\omega_k]P[\omega_k]} = \frac{p[x|\omega_j]P[\omega_j]}{p[x]}$$

where ω_j is the j -th class and x is the feature/observation vector

- **Decision Rule:** Choose class ω_j with highest $P[\omega_j|x]$?
 - Class is more “likely” given observation x

- **Terminology:**

$P[\omega_j]$: prior probability (of class ω_j)
 $P[\omega_j|x]$: posterior probability (of class ω_j given the observation x)
 $p[x|\omega_j]$: likelihood (probability of observation x given class ω_j)
 $p[x]$: normalization constant (does not affect decision)

What if misclassification costs are not the same?

Bayes Classifier: General Likelihood Ratio Test

- **Bayes Risk:** Minimize average “risk”
- For a two-class problem (classes C_1 and C_2)

$$\mathcal{R} = c_{21}p_1 \int_{\mathcal{X}_2} p_{\mathbf{x}}(\mathbf{x}|C_1)d\mathbf{x} + c_{12}p_2 \int_{\mathcal{X}_1} p_{\mathbf{x}}(\mathbf{x}|C_2)d\mathbf{x}$$

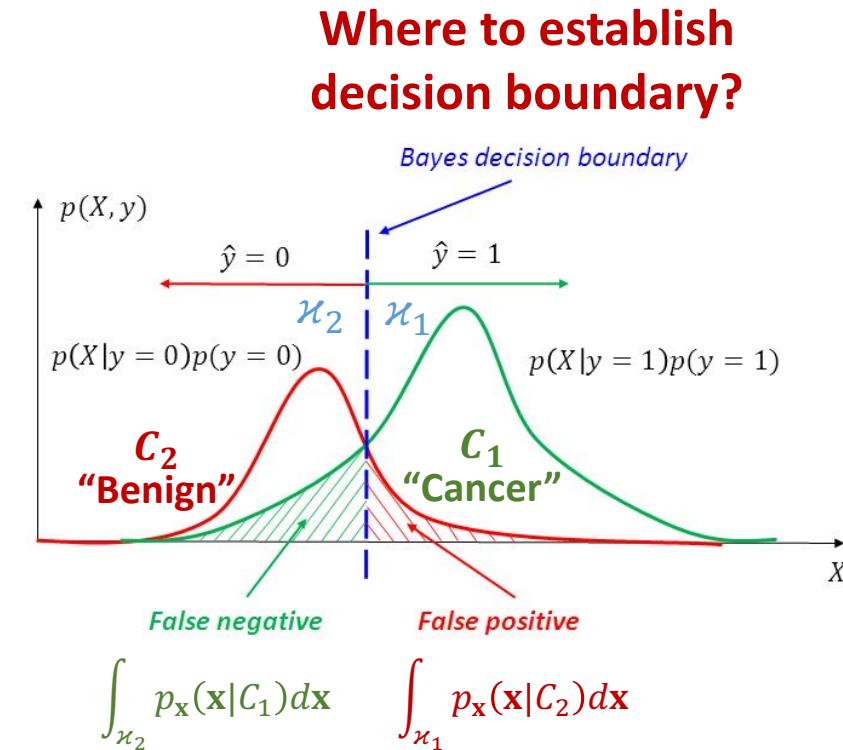
p_i : prior prob that $\mathbf{x}$ is from subspace $\mathcal{X}_i$

$$p_1 + p_2 = 1$$

c_{ij} : cost of deciding in favor of C_i when C_j is true

- Define: $\Lambda(\mathbf{x}) = \frac{p_{\mathbf{x}}(\mathbf{x}|C_1)}{p_{\mathbf{x}}(\mathbf{x}|C_2)}$ ← Likelihood ratio
- $\xi = \frac{p_2(c_{12}-c_{22})}{p_1(c_{21}-c_{11})}$ ← Threshold

- **Bayes Classifier:** If likelihood ratio $\Lambda(\mathbf{x}) > \xi$, assign $\mathbf{x}$ to class C_1 . Otherwise, assign it to class C_2 .



Bayes Classifier: Example

- **Clinical Problem:** Decide if a patient has a particular medical condition on the basis of an imperfect test!
- **Nomenclature:**
 - True-negative rate $P(-|\neg COND)$ of a test is called its SPECIFICITY
 - True-positive rate $P(+|COND)$ of a test is called its SENSITIVITY
- **Problem:**
 - Population of 10,000 with a 1% prevalence for condition
 - Test has 98% specificity and 90% sensitivity
 - Test result comes out POSITIVE
 - What is the probability that patient has condition?

Bayes Classifier: Example Solution

- **Applying Bayes Rule:**

$$\begin{aligned} P[COND|+] &= \frac{p[+|COND] P[COND]}{p[+]} \\ &= \frac{p[+|COND] P[COND]}{p[+|COND]P[COND]+p[+|\neg COND] P[\neg COND]} \\ &= \frac{0.90 \times 0.01}{0.90 \times 0.01 + (1-0.98) \times 0.99} \\ &= 0.3125 \end{aligned}$$

Naïve-Bayes Classifier

- **Pre-requisite for Bayes Classifier:** $p_{\mathbf{X}}(\mathbf{X}|C_i) = p_{\mathbf{X}}(X_1, X_2, \dots, X_m|C_i)$
- **Difficulty:** Learning this joint distribution as the size of the input vector grows
- Applying the “**chain rule**” repeatedly:

$$\begin{aligned}
 p_{\mathbf{X}}(X_1, X_2, \dots, X_m|C_i) &= p_{\mathbf{X}}(X_1|C_i)p_{\mathbf{X}}(X_2, \dots, X_m|C_i, X_1) \\
 &= p_{\mathbf{X}}(X_1|C_i)p_{\mathbf{X}}(X_2|C_i, X_1)p_{\mathbf{X}}(X_3, \dots, X_m|C_i, X_1, X_2) \\
 &\quad \dots
 \end{aligned}$$

$$= p_{\mathbf{X}}(X_1|C_i)p_{\mathbf{X}}(X_2|C_i, X_1)p_{\mathbf{X}}(X_3|C_i, X_1, X_2) \dots p_{\mathbf{X}}(X_m|C_i, X_1, X_2, X_3, \dots, X_{m-1})$$

- **Naïve Bayes Assumption:** All input attributes are conditionally independent!

$$\begin{aligned}
 p_{\mathbf{X}}(X_1, X_2, \dots, X_m|C_i) &= p_X(X_1|C_i)p_X(X_2|C_i, X_1)p_X(X_3|C_i, X_1, X_2) \dots p_X(X_m|C_i, X_1, X_2, X_3, \dots, X_{m-1}) \\
 &= p_X(X_1|C_i)p_X(X_2|C_i)p_X(X_3|C_i) \dots p_X(X_m|C_i) \\
 &= \prod_{i=1}^m p_{\mathbf{X}}(X_i|C_i)
 \end{aligned}$$

Symptoms for Bronchitis:

Cough; Mucus; Fatigue;
Shortness of breath;
Slight fever and chills ...

Assuming binary variables, compare #
of parameters in the two distributions!

More importantly, effective in practice!